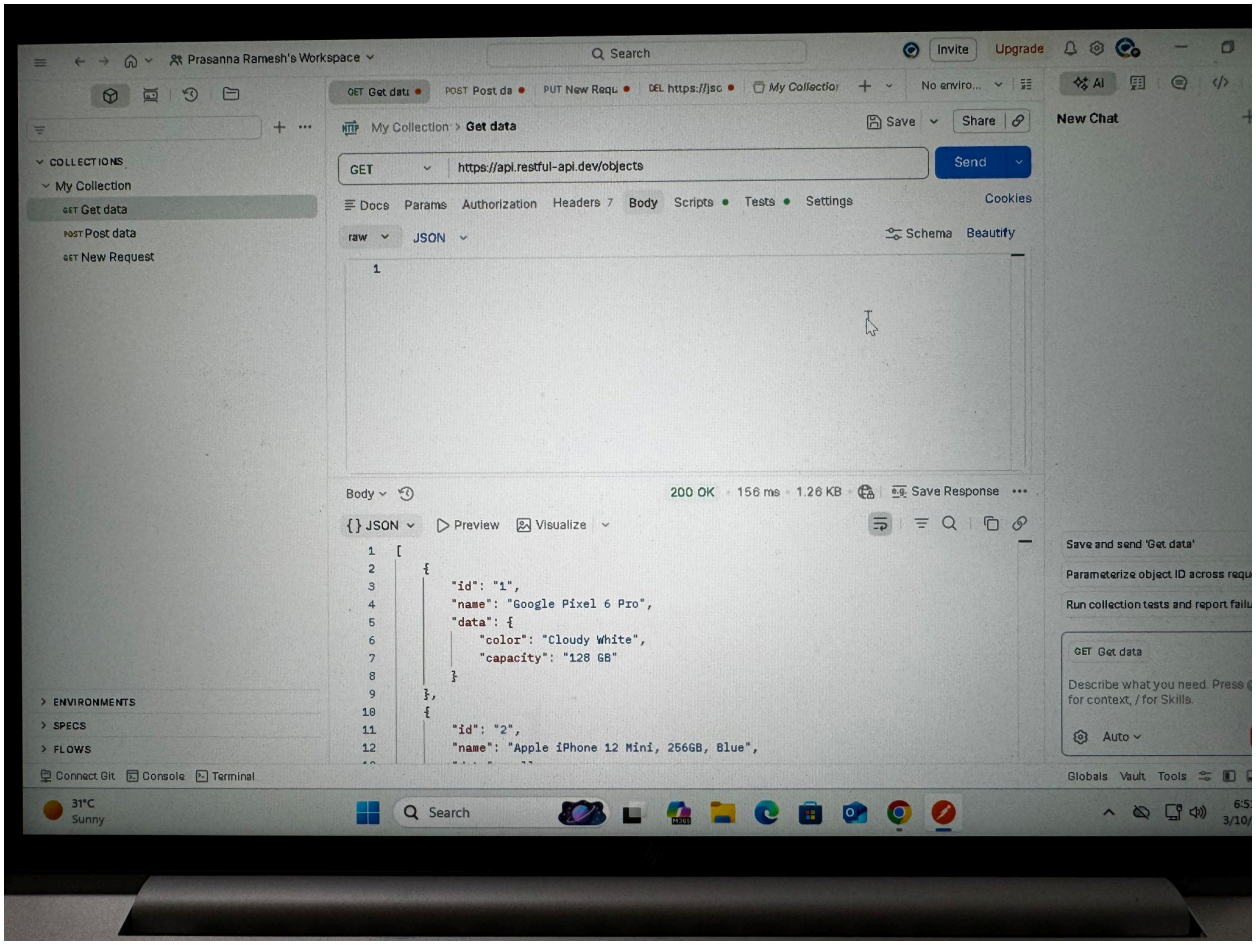
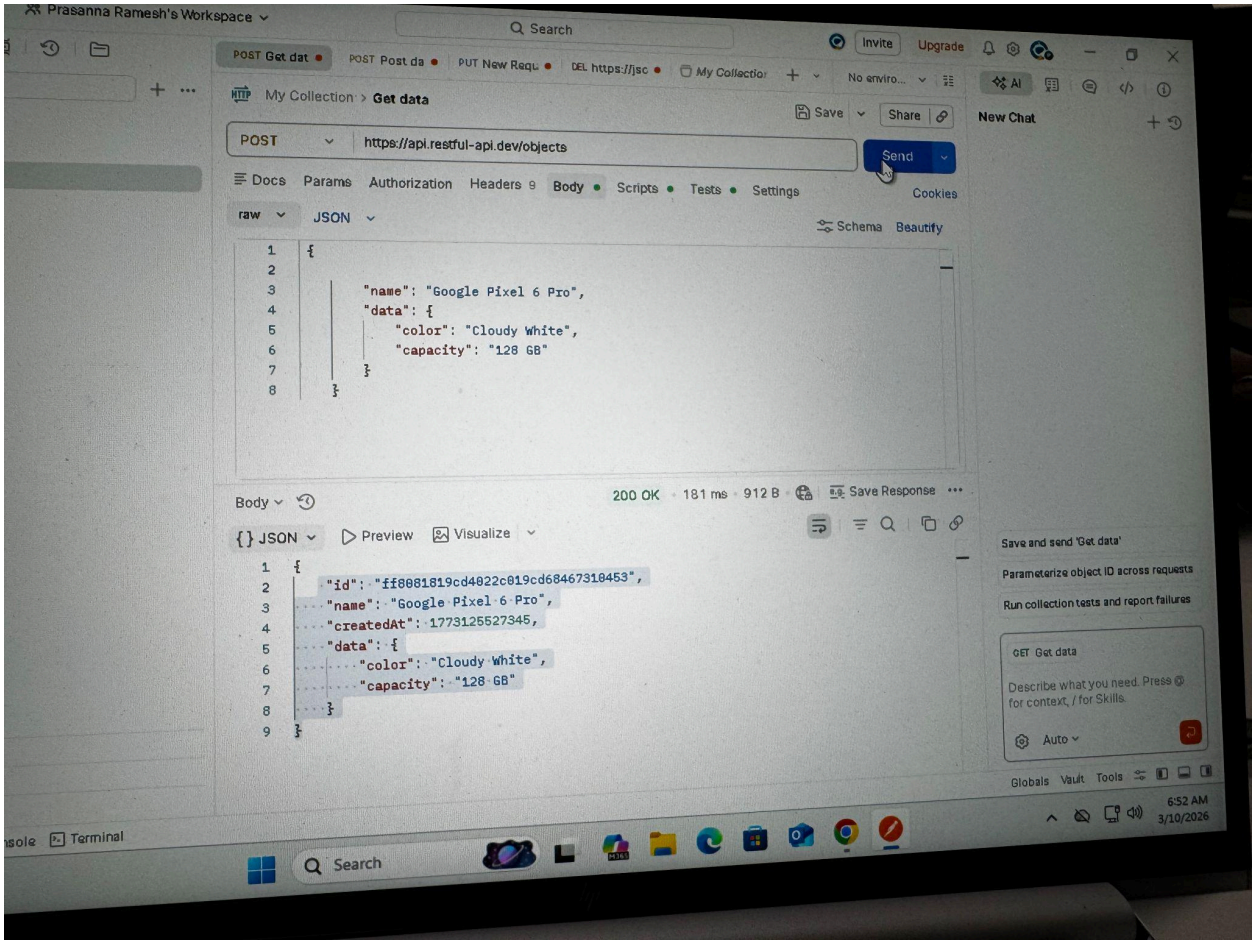
ACTIVITY I – Postman & HTTP Methods

Postman is a tool that lets developers interact with APIs directly. You can send requests, see how the server responds, and test your backend code without building a full front-end.

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New Chat
Save and send 'Get data'
Parameterize object ID across requests
Run collection tests and report failures
GET Get data
Describe what you need. Press @ for context, / for Skills.
Auto
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Search
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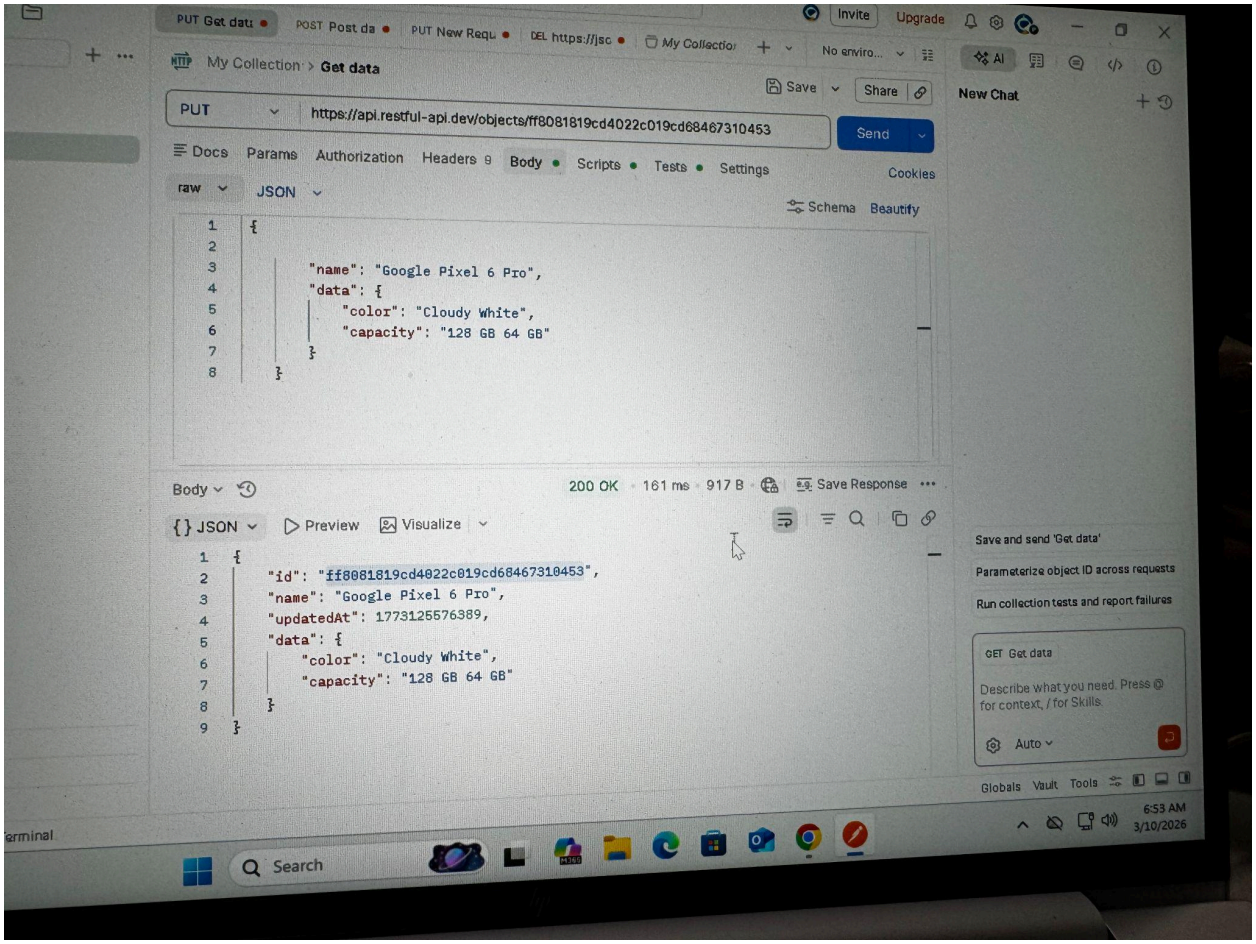
3/10/2026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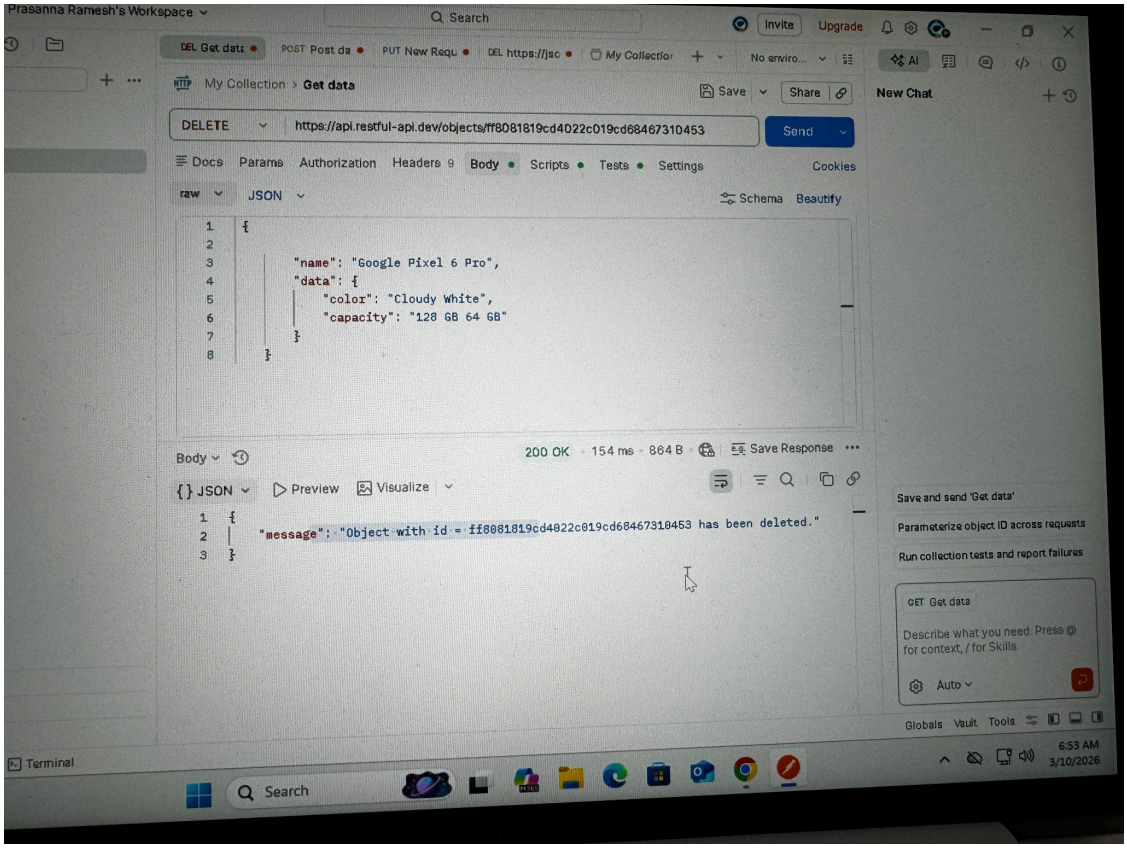

The Four Main HTTP Methods

Method	Purpose	Real-world Example	How to Use in Postman
GET	Retrieve data from a server	Checking your Instagram feed	Open Postman → Select GET → Enter URL → Click Send → View response
POST	Create a new resource on the server	Posting a tweet	Select POST → Go to Body → raw → JSON → Enter data → Send
PUT	Update an existing resource	Editing your profile bio	Select PUT → Body → raw → JSON → Enter updated data → Send
DELETE	Remove a resource	Deleting a photo	Select DELETE → Enter URL of the item → Send → Check status (200 OK / 204 No Content)

ACTIVITY II – GitHub Commands

GitHub is an online platform for storing code and tracking all changes. Commands help manage your local and remote repositories.

Core Concepts

- **Repository (Repo):** A folder that stores your project and its history.
- **Commit:** A save point in your project with a description.
- **Remote:** The version of your project stored online.

Essential Commands

Command	What it does	Analogy
`git init`	Turns a folder into a Git repository	Creating a notebook for your project
`git clone [URL]`	Copies an existing repo to your computer	Getting a friend's notebook
`git add [file]`	Stages files to be saved	Packing files into a box
`git commit -m "message"`	Saves changes permanently	Sealing the box with a label
`git push`	Uploads changes to GitHub	Sending the box online
`git pull`	Downloads latest changes from GitHub	Receiving updates from the cloud
`git branch [name]`	Creates a new branch	Making a side notebook for experiments
`git checkout [name]`	Switches branches	Opening another notebook
`git merge [name]`	Merges changes from one branch to another	Copying updates into main notebook
`git status`	Shows unsaved changes	Peeking inside your notebook

Tip: A typical workflow is: `git pull → edit → git add → git commit → git push`.

ACTIVITY III – Software Development Methodologies

1. Waterfall (Linear Approach)

- Follows a strict sequence: Requirements → Design → Implementation → Testing → Deployment → Maintenance.
- Best for short, simple projects or highly regulated environments.
- Feedback is only received at the end.
- Flexibility is very low.

2. Agile (Iterative / Sprints)

- Uses short iterations called **sprints** (2–4 weeks) with regular feedback.
- Daily stand-ups, sprint reviews, and retrospectives help improve the process.
- Core values:
    - Individuals & interactions over processes
    - Working software over documentation
    - Customer collaboration over contracts
    - Responding to change over following a strict plan
- Flexibility is high, feedback after every sprint.

3. DevOps (Continuous & Automated)

- Combines development and IT operations for **speed and reliability**.
- Workflow: Plan → Code → Build → Test → Release → Deploy → Operate → Monitor → back to Plan
- Key practices:
    - **CI (Continuous Integration):** Merge code multiple times a day

    - **CD (Continuous Deployment):** Automatically release tested changes
    - **IaC (Infrastructure as Code):** Manage servers with config files
- Feedback is continuous and real-time.
- Flexibility is extremely high.

Feature	Waterfall	Agile	DevOps
Structure	Linear	Iterative / Sprints	Continuous
Flexibility	Very Low	High	Extremely High
Feedback	End only	After each sprint	Continuous
Delivery	One big release	Frequent small releases	Continuous
Team Focus	Siloed	Collaborative	Integrated (Dev + Ops)